

TableCast: A Smart Surface Projector

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Scope

TableCast is designed to improve the experience of people learning new recipes in the kitchen by creating a projector that displays interactive steps onto a kitchen countertop. A user will be able to select a recipe on their laptop. Then, each step of the recipe will be demonstrated on the counter top so that the user can follow along using their ingredients.

Currently, most people watch YouTube videos or read articles online while they are cooking. There are a few problems with this. In many instances, fingers become messy from cooking. In order to look at the recipe again, you must constantly wash your hands or risk your device also becoming messy. This process is frustrating and unsanitary. Secondly, some people have trouble following the steps due to the differences between the instructor's tools/setup and one's own. There is no guide on how to cook something in one's own kitchen.

In our project, we aim to solve both issues. With our device, each step will be shown at the pace of the user's cooking, and recipe instructions will adapt to the user's own environment.

- 1) The user will install the projector in their kitchen so that it can project onto a flat surfaced countertop.
- 2) The user can open an app on their laptop to choose which recipe they want. They won't have to use their laptop again after this.
- 3) A projection of the ingredients will display on the countertop. Users must gather ingredients and put them in the correct spots.
- 4) The camera sends its video feed back to the laptop, which will use computer vision algorithms to detect if the user has all the ingredients and that they are in the right spot. If not, additional instructions will be given to the user.
- 5) If everything is detected correctly, the projection will display images on the countertop to show the first step of the recipe. When the user completes the step, the program will move onto the next step. As this all happens, all objects are tracked and each projected demonstration is specific to the user's real-time layout. Repeat until done.

Requirements

List of Requirements:

- Detect & identify user, ingredients and utensils quickly and accurately
- System should match the pace of the user
- System should be able to precisely build digital map of kitchen counter
- System program will run through a web application
- System should utilize depth sensors for object detection
- Full system is one unit that contains the projector, camera and additional sensors

Users: People who cook in the kitchen. More specifically, those who have trouble making recipes without instructions and/or new to cooking.

Key Technical Challenges:

- Identifying the user
- Identifying ingredients and utensils
- Tracking ingredients and utensils as the recipe is made
- Tracking user and identifying user gestures/motions
- Adapting projector display to various counter sizes

Technical Challenge Proposals:

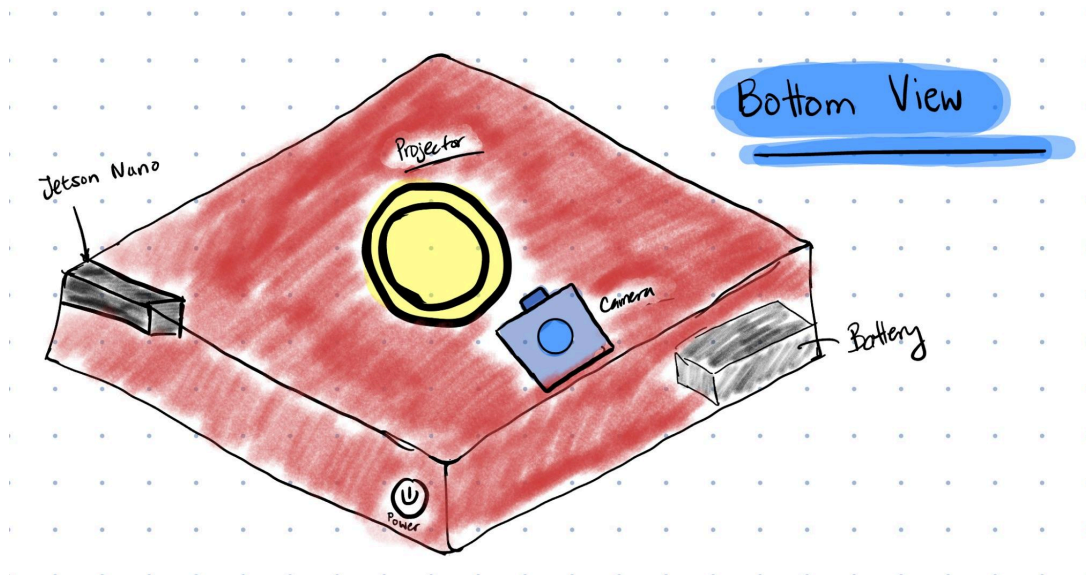
- Machine learning algorithms to help with object and user detection

Unknowns:

- Computing power required for our computer vision algorithms
- How will the camera and program will communicate
- Which products to use for each of components (i.e. projector, camera, sensors)

Implementation

Our project will be a single product built from scratch known as TableCast. It will contain a projector, a camera, the main computing board, and additional sensors as listed in the unknown subsection in the requirements category. TableCast will be installed such that it has a bird's eye view of the kitchen counter. Since we are building TableCast from scratch, we plan to use a Jetson Nano for managing power and controlling all the components. The camera and associated computer vision aspects will be conducted from the Nano itself.



Testing

We will invite other students to use our product. We will be measuring how long it takes for the test subject to complete the recipe and how accurate the recipe is. Accuracy will be measured through a rank system where the test subject will score their satisfaction and taste of the recipe they made. We will also be measuring the battery life compared to how long the test subject takes to cook. This will help us determine how long TableCast can operate for any given recipe.

MVP

Our MVP will be a smart device that includes the projects and camera. It should be able to guide the user through at least one predetermined recipe from our catalog. The user should be able to follow the recipe using the directions and instructions as displayed on the table surface. Our MVP does not include object identification and will rely on object detection to guide the user through the recipe. The product should be operable from a laptop.